

Rates of Change and Rectilinear Motion

ANSWER KEY



Position, velocity, and acceleration

- 1 A particle moves along a line with position $s(t) = t^3 - 6t^2 + 9t$ (metres, seconds).
- a Find the velocity function $v(t)$. $v(t) = 3t^2 - 12t + 9$.
- b Find the acceleration function $a(t)$. $a(t) = 6t - 12$.
- c Find $v(2)$ and $a(2)$. $v(2) = 12 - 24 + 9 = -3$ m/s; $a(2) = 12 - 12 = 0$ m/s².
- 2 For the particle above, find all times $t \geq 0$ when the particle is at rest ($v(t) = 0$).
- $3t^2 - 12t + 9 = 0 \Rightarrow t^2 - 4t + 3 = 0 \Rightarrow (t - 1)(t - 3) = 0$: $t = 1$ and $t = 3$.
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Speeding up and slowing down

- 3 Using the particle from above, determine whether the particle is speeding up or slowing down at $t = 2$ (compare the signs of $v(2)$ and $a(2)$).
- $v(2) = -3$ (negative) and $a(2) = 0$. Since $a(2) = 0$, the particle is neither speeding up nor slowing down at that instant — it is at a local extremum of speed.
- 4 Determine whether the particle is speeding up or slowing down at $t = 0$.
- $v(0) = 9 > 0$; $a(0) = -12 < 0$. Since v and a have opposite signs, the particle is slowing down at $t = 0$.
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Total distance vs. displacement

- 5 For the particle above, find its displacement over $[0, 3]$.
- Displacement $= s(3) - s(0) = (27 - 54 + 27) - 0 = 0$ m.
- 6 Using the fact that the particle changes direction at $t = 1$ (found earlier), find the total distance traveled over $[0, 3]$.
- $s(0) = 0$, $s(1) = 1 - 6 + 9 = 4$, $s(3) = 0$. Distance
- $= |s(1) - s(0)| + |s(3) - s(1)| = |4 - 0| + |0 - 4| = 4 + 4 = 8$ m.
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Related rates

- 7 A spherical balloon is inflated so its radius grows at 2 cm/s. Find the rate of change of its volume when $r = 6$ cm, using $V = \frac{4}{3}\pi r^3$.

$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt} = 4\pi(36)(2) = 288\pi \approx 904.8 \text{ cm}^3/\text{s}.$$

- 8 A 13 m ladder leans against a wall. The base slides away from the wall at 3 m/s. Find how fast the top of the ladder is sliding down when the base is 5 m from the wall.

$x^2 + y^2 = 169$: $2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$. At $x = 5$: $y = 12$. $5(3) + 12\frac{dy}{dt} = 0$, so $\frac{dy}{dt} = -\frac{15}{12} = -1.25$ m/s (sliding down at 1.25 m/s).

Motion from a velocity graph

- 9 A particle's velocity is $v(t) = t^2 - 4t$ (m/s) for $t \geq 0$. Find the times when the particle changes direction.

$v(t) = t(t - 4) = 0$: $t = 0$ and $t = 4$. The particle changes direction at $t = 4$ (velocity changes sign from negative to positive); $t = 0$ is just the start.

- 10 For the same particle, find the acceleration at $t = 1$ and state whether the particle is speeding up or slowing down at that instant.

$a(t) = 2t - 4$; $a(1) = -2$. $v(1) = 1 - 4 = -3$ (negative). Since $v(1)$ and $a(1)$ have the same sign (both negative), the particle is speeding up at $t = 1$.

Visualizing position, velocity, and acceleration together

- 11 The graph shows the position function $s(t) = t^3 - 6t^2 + 9t$ from the first problem. Use the graph to identify the interval(s) where the particle is moving to the left (i.e. s is decreasing).

The graph decreases on $(1, 3)$ (between the two points found earlier where $v(t) = 0$), so the particle moves to the left (negative direction) during that interval.

An applied related-rates capstone

- 12 A plane flies at a constant altitude of 3 miles, moving horizontally at 500 mph, directly away from a radar station on the ground. Find the rate at which the distance between the plane and the radar station is increasing when the plane is 4 miles horizontally from the station (directly above a point 4 miles from the station).

Let x = horizontal distance, s = distance from radar (with height fixed at 3): $s^2 = x^2 + 9$. $2s\frac{ds}{dt} = 2x\frac{dx}{dt}$. At $x = 4$: $s = \sqrt{16 + 9} = 5$. $5\frac{ds}{dt} = 4(500) = 2000$, so $\frac{ds}{dt} = 400$ mph.

Motion with a velocity given as a formula

- 13** A particle's velocity is $v(t) = 6t - t^2$ (m/s) for $0 \leq t \leq 8$. Find the particle's position function $s(t)$ given $s(0) = 0$, then find its displacement over $[0, 6]$.

$$s(t) = \int v(t) dt = 3t^2 - \frac{t^3}{3} + C. \text{ Using } s(0) = 0: C = 0. s(t) = 3t^2 - \frac{t^3}{3}. \text{ Displacement} \\ = s(6) - s(0) = (108 - 72) - 0 = 36 \text{ m.}$$

A capstone rectilinear motion problem

- 14** A rocket's height is $h(t) = -5t^2 + 40t$ (meters, seconds) after launch. Find its maximum height, and the time at which it returns to the ground.

$$h'(t) = -10t + 40 = 0 \text{ gives } t = 4 \text{ (time of maximum height). } h(4) = -80 + 160 = 80 \text{ m. Ground return:} \\ h(t) = 0 \text{ gives } -5t(t - 8) = 0: t = 0 \text{ or } t = 8 \text{ s.}$$

Multiple choice — easy 15

- 15** A particle's position is $s(t) = -5t^2 + 2t - 4$ (meters, seconds). Find its velocity $v(4)$.

- a. -18
- b. -36
- c. -40
- d. -38 ✓

$$v(t) = -10t + 2, \text{ so } v(4) = -10(4) + 2 = -38 \text{ m/s.}$$

- 16** A particle's position is $s(t) = -6t^2 - 3t + 9$ (meters, seconds). Find its velocity $v(5)$.

- a. -65
- b. -63 ✓
- c. -61
- d. -33

$$v(t) = -12t - 3, \text{ so } v(5) = -12(5) - 3 = -63 \text{ m/s.}$$

17 A particle's position is $s(t) = 6t^2 + 9t - 5$ (meters, seconds). Find its velocity $v(3)$.

a. 27

b. 45 ✓

c. 47

d. 43

$v(t) = 12t + 9$, so $v(3) = 12(3) + 9 = 45$ m/s.

18 A particle's position is $s(t) = -2t^2 + t + 8$ (meters, seconds). Find its velocity $v(4)$.

a. -13

b. -17

c. -7

d. -15 ✓

$v(t) = -4t + 1$, so $v(4) = -4(4) + 1 = -15$ m/s.

19 A particle's position is $s(t) = t^2 - 5t - 9$ (meters, seconds). Find its velocity $v(3)$.

a. -2

b. -1

c. 3

d. 1 ✓

$v(t) = 2t - 5$, so $v(3) = 2(3) - 5 = 1$ m/s.

20 A particle's position is $s(t) = -6t^2 + 7t - 4$ (meters, seconds). Find its velocity $v(0)$.

a. 9

b. 8

c. 7 ✓

d. 5

$v(t) = -12t + 7$, so $v(0) = -12(0) + 7 = 7$ m/s.

21 A particle's position is $s(t) = 3t^2 - 8t + 3$ (meters, seconds). Find its velocity $v(4)$.

a. 14

b. 4

c. 16 ✓

d. 18

$v(t) = 6t - 8$, so $v(4) = 6(4) - 8 = 16$ m/s.

22 A particle's position is $s(t) = 4t^2 - 5t$ (meters, seconds). Find its velocity $v(4)$.

a. 11

b. 25

c. 27 ✓

d. 29

$v(t) = 8t - 5$, so $v(4) = 8(4) - 5 = 27$ m/s.

23 A particle's position is $s(t) = -6t^2 + 7t - 9$ (meters, seconds). Find its velocity $v(3)$.

a. -27

b. -31

c. -29 ✓

d. -11

$v(t) = -12t + 7$, so $v(3) = -12(3) + 7 = -29$ m/s.

24 A particle's position is $s(t) = -5t^2 - 3t + 6$ (meters, seconds). Find its velocity $v(2)$.

a. -23 ✓

b. -25

c. -13

d. -21

$v(t) = -10t - 3$, so $v(2) = -10(2) - 3 = -23$ m/s.

25 A particle's position is $s(t) = t^2 + 9t + 4$ (meters, seconds). Find its velocity $v(5)$.

a. 19 ✓

b. 14

c. 17

d. 21

$v(t) = 2t + 9$, so $v(5) = 2(5) + 9 = 19$ m/s.

26 A particle's position is $s(t) = 4t^2 + 9t$ (meters, seconds). Find its velocity $v(2)$.

a. 25 ✓

b. 23

c. 27

d. 17

$v(t) = 8t + 9$, so $v(2) = 8(2) + 9 = 25$ m/s.

27 A particle's position is $s(t) = 5t^2 + 5t - 7$ (meters, seconds). Find its velocity $v(3)$.

a. 35 ✓

b. 20

c. 37

d. 33

$v(t) = 10t + 5$, so $v(3) = 10(3) + 5 = 35$ m/s.

28 A particle's position is $s(t) = 3t^2 - 4t - 8$ (meters, seconds). Find its velocity $v(1)$.

a. 4

b. -1

c. 2 ✓

d. 0

$v(t) = 6t - 4$, so $v(1) = 6(1) - 4 = 2$ m/s.

29 A particle's position is $s(t) = -5t^2 + 6t - 8$ (meters, seconds). Find its velocity $v(1)$.

a. -4 ✓

b. 1

c. -2

d. -6

$v(t) = -10t + 6$, so $v(1) = -10(1) + 6 = -4$ m/s.

Multiple choice — medium 15

30 A particle's position is $s(t) = t^3 + 6t$ (meters, seconds). Find its acceleration $a(3)$.

a. 12

b. 18 ✓

c. 24

d. 9

$v(t) = 3t^2 + 6$; $a(t) = 6t$, so $a(3) = 18 = 18$ m/s².

31 A particle's position is $s(t) = t^3 + t^2 - 2t$ (meters, seconds). Find its acceleration $a(2)$.

a. 20

b. 14 ✓

c. 7

d. 8

$$v(t) = 3t^2 + 2t - 2; a(t) = 6t + 2, \text{ so } a(2) = 12 + 2 = 14 \text{ m/s}^2.$$

32 A particle's position is $s(t) = t^3 - 4t^2 + 3t$ (meters, seconds). Find its acceleration $a(1)$.

a. -1

b. -2 ✓

c. 4

d. -8

$$v(t) = 3t^2 - 8t + 3; a(t) = 6t - 8, \text{ so } a(1) = 6 - 8 = -2 \text{ m/s}^2.$$

33 A particle's position is $s(t) = t^3 - 6t^2 - 8t$ (meters, seconds). Find its acceleration $a(5)$.

a. 24

b. 9

c. 12

d. 18 ✓

$$v(t) = 3t^2 - 12t - 8; a(t) = 6t - 12, \text{ so } a(5) = 30 - 12 = 18 \text{ m/s}^2.$$

34 A particle's position is $s(t) = t^3 - 2t^2 + 8t$ (meters, seconds). Find its acceleration $a(3)$.

a. 20

b. 8

c. 7

d. 14 ✓

$$v(t) = 3t^2 - 4t + 8; a(t) = 6t - 4, \text{ so } a(3) = 18 - 4 = 14 \text{ m/s}^2.$$

35 A particle's position is $s(t) = t^3 - 2t^2 - 2t$ (meters, seconds). Find its acceleration $a(2)$.

a. 8 ✓

b. 4

c. 14

d. 2

$$v(t) = 3t^2 - 4t - 2; a(t) = 6t - 4, \text{ so } a(2) = 12 - 4 = 8 \text{ m/s}^2.$$

36 A particle's position is $s(t) = t^3 - 4t^2 - 5t$ (meters, seconds). Find its acceleration $a(3)$.

a. 5

b. 4

c. 16

d. 10 ✓

$$v(t) = 3t^2 - 8t - 5; a(t) = 6t - 8, \text{ so } a(3) = 18 - 8 = 10 \text{ m/s}^2.$$

37 A particle's position is $s(t) = t^3 + t^2 - 7t$ (meters, seconds). Find its acceleration $a(1)$.

a. 2

b. 4

c. 14

d. 8 ✓

$$v(t) = 3t^2 + 2t - 7; a(t) = 6t + 2, \text{ so } a(1) = 6 + 2 = 8 \text{ m/s}^2.$$

38 A particle's position is $s(t) = t^3 + 6t^2 + 6t$ (meters, seconds). Find its acceleration $a(4)$.

a. 36 ✓

b. 18

c. 30

d. 42

$$v(t) = 3t^2 + 12t + 6; a(t) = 6t + 12, \text{ so } a(4) = 24 + 12 = 36 \text{ m/s}^2.$$

39 A particle's position is $s(t) = t^3 - 4t$ (meters, seconds). Find its acceleration $a(0)$.

a. 5

b. -6

c. 0 ✓

d. 6

$$v(t) = 3t^2 - 4; a(t) = 6t, \text{ so } a(0) = 0 = 0 \text{ m/s}^2.$$

40 A particle's position is $s(t) = t^3 - t^2 - 5t$ (meters, seconds). Find its acceleration $a(5)$.

a. 22

b. 34

c. 28 ✓

d. 14

$$v(t) = 3t^2 - 2t - 5; a(t) = 6t - 2, \text{ so } a(5) = 30 - 2 = 28 \text{ m/s}^2.$$

41 A particle's position is $s(t) = t^3 + t^2 + 6t$ (meters, seconds). Find its acceleration $a(2)$.

a. 20

b. 8

c. 14 ✓

d. 7

$$v(t) = 3t^2 + 2t + 6; a(t) = 6t + 2, \text{ so } a(2) = 12 + 2 = 14 \text{ m/s}^2.$$

42 A particle's position is $s(t) = t^3 + 5t^2 - t$ (meters, seconds). Find its acceleration $a(2)$.

a. 28

b. 22 ✓

c. 16

d. 11

$$v(t) = 3t^2 + 10t - 1; a(t) = 6t + 10, \text{ so } a(2) = 12 + 10 = 22 \text{ m/s}^2.$$

43 A particle's position is $s(t) = t^3 + 6t^2 - t$ (meters, seconds). Find its acceleration $a(1)$.

a. 24

b. 18 ✓

c. 12

d. 9

$$v(t) = 3t^2 + 12t - 1; a(t) = 6t + 12, \text{ so } a(1) = 6 + 12 = 18 \text{ m/s}^2.$$

44 A particle's position is $s(t) = t^3 + 5t^2 + 2t$ (meters, seconds). Find its acceleration $a(3)$.

a. 28 ✓

b. 14

c. 22

d. 34

$$v(t) = 3t^2 + 10t + 2; a(t) = 6t + 10, \text{ so } a(3) = 18 + 10 = 28 \text{ m/s}^2.$$

Multiple choice — hard 15

- 45 A particle's velocity is $v(t) = t^2 - t + 4$ (m/s). Determine whether the particle is speeding up or slowing down at $t = 2$.

a. $v(2) = 6$ and $a(2) = 3$ — same sign, so the particle is slowing down.

b. $v(2) = 6$ and $a(2) = 3$ — opposite signs, so the particle is speeding up.

c. $v(2) = 6$ and $a(2) = 3$ — same sign, so the particle is speeding up. ✓

d. $v(2) = 6$ and $a(2) = 3$ — opposite signs, so the particle is slowing down.

$v(t) = t^2 - t + 4$, $a(t) = 2t - 1$: $v(2) = 6$ and $a(2) = 3$ — same sign, so the particle is speeding up.

- 46 A particle's velocity is $v(t) = t^2 + 5t - 4$ (m/s). Determine whether the particle is speeding up or slowing down at $t = 2$.

a. $v(2) = 10$ and $a(2) = 9$ — same sign, so the particle is slowing down.

b. $v(2) = 10$ and $a(2) = 9$ — same sign, so the particle is speeding up. ✓

c. $v(2) = 10$ and $a(2) = 9$ — opposite signs, so the particle is slowing down.

d. $v(2) = 10$ and $a(2) = 9$ — opposite signs, so the particle is speeding up.

$v(t) = t^2 + 5t - 4$, $a(t) = 2t + 5$: $v(2) = 10$ and $a(2) = 9$ — same sign, so the particle is speeding up.

- 47 A particle's velocity is $v(t) = t^2 + 4t + 2$ (m/s). Determine whether the particle is speeding up or slowing down at $t = 3$.

a. $v(3) = 23$ and $a(3) = 10$ — same sign, so the particle is speeding up. ✓

b. $v(3) = 23$ and $a(3) = 10$ — opposite signs, so the particle is speeding up.

c. $v(3) = 23$ and $a(3) = 10$ — same sign, so the particle is slowing down.

d. $v(3) = 23$ and $a(3) = 10$ — opposite signs, so the particle is slowing down.

$v(t) = t^2 + 4t + 2$, $a(t) = 2t + 4$: $v(3) = 23$ and $a(3) = 10$ — same sign, so the particle is speeding up.

- 48 A particle's velocity is $v(t) = t^2 - 2t + 3$ (m/s). Determine whether the particle is speeding up or slowing down at $t = 0$.

- a. $v(0) = 3$ and $a(0) = -2$ — opposite signs, so the particle is speeding up.
- b. $v(0) = 3$ and $a(0) = -2$ — opposite signs, so the particle is slowing down. ✓
- c. $v(0) = 3$ and $a(0) = -2$ — same sign, so the particle is slowing down.
- d. $v(0) = 3$ and $a(0) = -2$ — same sign, so the particle is speeding up.

$v(t) = t^2 - 2t + 3$, $a(t) = 2t - 2$: $v(0) = 3$ and $a(0) = -2$ — opposite signs, so the particle is slowing down.

- 49 A particle's velocity is $v(t) = t^2 - 4t + 6$ (m/s). Determine whether the particle is speeding up or slowing down at $t = 1$.

- a. $v(1) = 3$ and $a(1) = -2$ — opposite signs, so the particle is speeding up.
- b. $v(1) = 3$ and $a(1) = -2$ — same sign, so the particle is slowing down.
- c. $v(1) = 3$ and $a(1) = -2$ — opposite signs, so the particle is slowing down. ✓
- d. $v(1) = 3$ and $a(1) = -2$ — same sign, so the particle is speeding up.

$v(t) = t^2 - 4t + 6$, $a(t) = 2t - 4$: $v(1) = 3$ and $a(1) = -2$ — opposite signs, so the particle is slowing down.

- 50 A 17 m ladder leans against a wall. The base slides away from the wall at 1 m/s. Find how fast the top is sliding down when the base is 8 m from the wall.

- a. -1
- b. $-\frac{8}{15}$ ✓
- c. $\frac{15}{8}$
- d. $\frac{8}{15}$

$x^2 + y^2 = 17^2$: $2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$. At $x = 8$: $y = 15$. $(8)(1) + (15)\frac{dy}{dt} = 0$, so $\frac{dy}{dt} = \frac{-8}{15} = -\frac{8}{15}$ m/s.

- 51 A 10 m ladder leans against a wall. The base slides away from the wall at 5 m/s. Find how fast the top is sliding down when the base is 6 m from the wall.

- a. $\frac{20}{3}$
- b. -5
- c. $-\frac{15}{4}$ ✓
- d. $\frac{15}{4}$

$$x^2 + y^2 = 10^2: 2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0. \text{ At } x = 6: y = 8. (6)(5) + (8)\frac{dy}{dt} = 0, \text{ so } \frac{dy}{dt} = \frac{-30}{8} = -\frac{15}{4} \text{ m/s.}$$

- 52 A 5 m ladder leans against a wall. The base slides away from the wall at 4 m/s. Find how fast the top is sliding down when the base is 3 m from the wall.

- a. $\frac{16}{3}$
- b. -3 ✓
- c. -4
- d. 3

$$x^2 + y^2 = 5^2: 2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0. \text{ At } x = 3: y = 4. (3)(4) + (4)\frac{dy}{dt} = 0, \text{ so } \frac{dy}{dt} = \frac{-12}{4} = -3 \text{ m/s.}$$

- 53 A 15 m ladder leans against a wall. The base slides away from the wall at 3 m/s. Find how fast the top is sliding down when the base is 9 m from the wall.

- a. 4
- b. -3
- c. $-\frac{9}{4}$ ✓
- d. $\frac{9}{4}$

$$x^2 + y^2 = 15^2: 2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0. \text{ At } x = 9: y = 12. (9)(3) + (12)\frac{dy}{dt} = 0, \text{ so } \frac{dy}{dt} = \frac{-27}{12} = -\frac{9}{4} \text{ m/s.}$$

- 54 A 13 m ladder leans against a wall. The base slides away from the wall at 1 m/s. Find how fast the top is sliding down when the base is 5 m from the wall.

- a. -1
- b. $\frac{12}{5}$
- c. $-\frac{5}{12}$ ✓
- d. $\frac{5}{12}$

$x^2 + y^2 = 13^2$: $2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$. At $x = 5$: $y = 12$. $(5)(1) + (12)\frac{dy}{dt} = 0$, so $\frac{dy}{dt} = \frac{-5}{12} = -\frac{5}{12}$ m/s.

- 55 A particle's position is $s(t) = t^3 - 12t^2 + 45t$ (meters, seconds), which is at rest at $t = 3$ and $t = 5$. Find the total distance traveled over $[0, 5]$.

- a. 50
- b. 60
- c. 58 ✓
- d. 56

$s(0) = 0$, $s(3) = 54$, $s(5) = 50$. Distance = $|54 - 0| + |50 - 54| = 54 + 4 = 58$ m.

- 56 A particle's position is $s(t) = t^3 - 9t^2 + 15t$ (meters, seconds), which is at rest at $t = 1$ and $t = 5$. Find the total distance traveled over $[0, 5]$.

- a. 25
- b. 37
- c. 39 ✓
- d. 41

$s(0) = 0$, $s(1) = 7$, $s(5) = -25$. Distance = $|7 - 0| + |-25 - 7| = 7 + 32 = 39$ m.

- 57 A particle's position is $s(t) = t^3 - 15t^2 + 72t$ (meters, seconds), which is at rest at $t = 4$ and $t = 6$. Find the total distance traveled over $[0, 6]$.

a. 116 ✓

b. 114

c. 108

d. 118

$s(0) = 0, s(4) = 112, s(6) = 108$. Distance = $|112 - 0| + |108 - 112| = 112 + 4 = 116$ m.

- 58 A particle's position is $s(t) = t^3 - 9t^2 + 24t$ (meters, seconds), which is at rest at $t = 2$ and $t = 4$. Find the total distance traveled over $[0, 4]$.

a. 16

b. 24 ✓

c. 26

d. 22

$s(0) = 0, s(2) = 20, s(4) = 16$. Distance = $|20 - 0| + |16 - 20| = 20 + 4 = 24$ m.

- 59 A particle's position is $s(t) = t^3 - 15t^2 + 63t$ (meters, seconds), which is at rest at $t = 3$ and $t = 7$. Find the total distance traveled over $[0, 7]$.

a. 111

b. 113 ✓

c. 115

d. 49

$s(0) = 0, s(3) = 81, s(7) = 49$. Distance = $|81 - 0| + |49 - 81| = 81 + 32 = 113$ m.